Anish Banerjee

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http://AB271202.github.io/
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Education

2021 – 2025 **Indian Institute of Technology, Delhi** **CGPA:** 9.52/10 (3.8/4.0)
BTech in Computer Science and Engineering

2019 – 2021 **Bal Bhavan International School, Dwarka, Delhi** **Percentage:** 96.4%
Intermediate / +2

2009 – 2019 **Mount Carmel School, Dwarka, Delhi** **Percentage:** 96.2%
Matriculation

Research Internships

2024 **Institute of Quantum Computing (IQC)** **Title:** Unclonable Cryptography
UNIVERSITY OF WATERLOO *Guide: Prof. Ashwin Nayak*

- Studied Brodabent and Lord's paper [BL20] on Unclonable Encryption, gaining an understanding of the theoretical framework behind quantum encryption schemes and the impossibility of cloning quantum ciphertexts.
- Reviewed Tomamichel et al.'s paper [TFKW13] on the Monogamy of Entanglement Games, focusing on how entanglement properties affect quantum communication protocols and their applications in cryptographic tasks.
- Explored alternative methods to prove the security of conjugate encryption scheme in [BL20], aiming to extend it for multi-party encryption.

2023 **Center for Quantum Technologies (CQT)** **Title:** Quantum Advantage
NATIONAL UNIVERSITY OF SINGAPORE *Guide: Prof. Rahul Jain*

- Explored methods for demonstrating quantum advantage including [AGL+23] on noisy random circuit sampling, [BCM+21], [BKVV20] on cryptographic tests of quantumness, and [KLVY22] on compiling non-local games into a proof of quantumness protocol.
- Presented research insights, enhancing my communication skills and contributing to discussions. My training report, along with slides and notes, is available here.

Expository Writings

2023-2024 Prepared expository writings on various topics pertaining to quantum information which can be found here: https://ab271202.github.io/notes

Projects

2024-Present **BTech Project: Quantum Random Oracles** *Guide: Prof. Venkata Koppula*

- Investigated the random oracle model in quantum settings, comparing its impact on classical proofs: analyzed proof techniques in the QROM, including History-free Reductions [BDF+11], One-Way to Hiding [Unr14], and Compressed Oracles [Zha19].
- Proved the security of Nielsen's non-committing encryption construction [Nie02] in the quantum random oracle model.
- Analyzed quantum public key encryption constructions [Col23], [BGH+23], and explored minimal assumptions for achieving quantum CCA security.
- Attempted a construction of quantum CCA1 secure scheme from pseudorandom states.

2023-Present **BQP Verification** *Guide: Prof. Venkata Koppula and Prof. Rajendra Kumar*

- Explored interactive proofs for quantum computation through Prof. Thomas Vidick's course [FSMP].
- Analyzed Mahadev's pioneering work on classical verification of quantum computations [Mah23], gaining in-depth understanding of the protocol and construction of the function families; find my report here.

Workshops

2024 **Undergraduate School on Experimental Quantum Information Processing [USEQIP]**
INSTITUTE FOR QUANTUM COMPUTING, UNIVERSITY OF WATERLOO
Selected as one of 30 undergraduates globally for the week-long program on theoretical and experimental quantum information processing.

CryptoWorks21 [CWs21]
INSTITUTE FOR QUANTUM COMPUTING, UNIVERSITY OF WATERLOO
Participated in a workshop focused on quantum-safe cryptography and information security.

2023 **Quantum Camp [QCamp]**
CENTER FOR QUANTUM TECHNOLOGIES, NATIONAL UNIVERSITY OF SINGAPORE
Participated in a week-long intensive program on quantum computing and quantum technologies, involving lectures, industry visits and hands-on lab sessions.

Service

2024 **Teaching Assistantship**
[COL202] DISCRETE MATHEMATICS *Instructor: Prof. Venkata Koppula*
One of 12 undergraduates selected to serve as a TA. Led tutorial sessions for a group of students throughout the semester, offering personalized assistance with conceptual challenges, and providing feedback on proof-writing techniques.

Service (continued)

2023 **Academic Mentor**
[PYL101] ELECTROMAGNETISM & QUANTUM MECHANICS *Instructor: Prof. Vikrant Saxena*
Responsible for conducting personalized doubt and revision sessions for first-year students.

Scholastic Achievements

2024 **Undergraduate Research Award**
Selected as one of 30 undergraduates worldwide for a summer school and research internship at the Institute for Quantum Computing, University of Waterloo.

2022 **IIT Delhi Semester Merit Award**
Awarded for being among the top 7% of students in both semesters I and II.

Endowment Scholarship
Awarded for being among the top 15 students in the 2021 entry batch at IIT Delhi.

2021 **Joint Entrance Exam (JEE) Main**
Secured AIR 40 out of 1,000,000 candidates who appeared for the exam.

Joint Entrance Exam (JEE) Advanced
Secured AIR 373 out of 200,000 candidates who appeared for the exam.

KVPY Fellowship
Conferred by DST GoI, secured AIR 134 out of 100,000 candidates who appeared for the exam.

IOQP (Physics), IOQC (Chemistry)
Secured a place in the top 1% in these olympiads among 200,000 candidates.

2019 **NTSE Scholarship**
Awarded by NCERT for ranking in the top 0.1% of students among over 1 million applicants.

Select Courses Undertaken

Computer Science	Lattices in Computer Science, Basic Information Theory, Complexity Theory, Cryptography and Computer Security
Mathematics	Probability and Stochastic Processes, Algebra
Quantum Information	Applied Quantum Mechanics, Quantum Algorithms, Quantum Cryptography, Quantum Complexity Theory

Extra-Curricular Activities and Volunteering

2023-2024 **Events Executive: Physics and Astronomy Club**
Involved in organizing several events related to physics and astronomy including Cosmicon fest, Physics Olympiad, Quantum Computing Workshop etc.

2022-2023 **Volunteering: National Service Scheme**
Volunteered for the Vidya teaching project under NSS, IIT Delhi, providing educational support to students from economically disadvantaged backgrounds. Awarded a trophy for "*Significant Contribution to NSS, IITD*."